

WEST HAWAII BUOY 51003, USA

WMO: 992610

Lat: 19.175N

Lon: 160.625W

Elev: 0

StdP: 101.33

Time Zone: -11.00 (W11)

Period: 94-18

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
3	21.7	22.2	N/A	N/A	N/A	N/A	N/A	N/A	13.7	23.2	12.0	23.3	7.0	40	0.563

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
9	1.5	28.2	N/A	27.9	N/A	27.7	N/A	N/A	N/A	N/A	N/A	N/A	N/A	5.2	70

Dehumidification DP/MCDB and HR

0.4%			1%			2%			0.4%		1%		2%		Extreme Max WB
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a) N/A	(b) N/A	(c) N/A	(d) N/A	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(n) N/A	(o) N/A	

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		Extreme Max WB	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	(q)	(r)
11.1	10.3	9.7	20.2	28.4	0.8	0.5	19.6	28.8	19.1	29.1	18.7	29.4	18.1	29.8	N/A	N/A
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	25.1	24.0	23.9	23.7	24.1	24.7	25.4	25.9	26.4	26.4	26.4	25.6	24.8
	DBStd	1.23	0.77	0.78	0.74	0.74	0.71	0.64	0.67	0.64	0.73	0.67	0.72	0.78
	HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD18.3	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDD10.0	5513	433	388	426	423	455	461	492	508	493	507	469	458
	CDD18.3	2471	175	155	167	173	197	211	234	249	243	249	219	199
	CDH23.3	17482	651	527	521	742	1146	1620	2047	2399	2393	2414	1807	1215
	CDH26.7	608	0	0	0	0	1	4	45	149	217	164	28	1
Wind	WSAvg	6.2	6.0	6.3	6.9	6.8	5.9	5.9	6.2	6.0	5.4	5.9	6.8	6.7
Precipitation	PrecAvg	437	38	51	51	26	15	12	17	27	28	52	60	60
	PrecMax	879	160	153	231	125	67	93	56	128	137	130	214	274
	PrecMin	203	4	2	3	4	4	1	3	4	3	4	2	6
	PrecStd	168	35	48	57	26	15	16	13	27	27	34	50	63
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	26.3	25.8	25.7	25.8	26.9	27.1	27.9	28.5	29.0	28.0	27.6	26.9
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	2%	DB	25.8	25.7	25.3	25.7	26.5	26.9	27.5	28.0	28.5	27.9	27.3	26.5
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	5%	DB	25.4	25.4	25.2	25.6	26.2	26.7	27.2	27.7	28.0	27.7	27.0	26.2
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	10%	DB	25.2	25.1	25.0	25.3	25.8	26.4	26.9	27.4	27.7	27.5	26.8	25.9
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	2%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	5%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	10%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Mean Daily Temperature Range	5% DB	MDBR	1.5	1.5	1.5	1.4	1.4	1.4	1.4	1.3	1.5	1.4	1.4	1.3
		MCDBR	1.6	1.5	1.5	1.5	1.5	1.5	1.5	1.3	1.4	1.7	1.4	1.5
		MCWBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	5% WB	MCDBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCWBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Clear-Sky Solar Irradiance	taub	0.360	0.364	0.375	0.410	0.407	0.398	0.400	0.398	0.387	0.381	0.374	0.361	
	taud	2.601	2.588	2.562	2.436	2.468	2.492	2.501	2.506	2.525	2.590	2.595	2.616	
	Ebn at Noon	924	936	934	897	889	888	888	896	908	908	901	911	
	Edh at Noon	92	97	103	118	113	109	109	109	105	96	92	88	
All-Sky Solar Radiation	RadAvg	4.75	5.22	5.79	6.33	6.68	6.75	6.74	6.62	6.20	5.36	4.61	4.34	
	RadStd	0.21	0.31	0.59	0.45	0.41	0.25	0.20	0.21	0.25	0.22	0.19	0.28	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (0 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page